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Business Understanding and Requirements

Life Insurance Policy Retention

Week 2

The retention team's decision

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Fictional insurer

Ana's premium debit failed after she changed bank accounts. Ben missed a premium after a drop in household income. Illustrative capacity: 100 policy reviews this month. Confirm contact capacity separately.

Driving question

Which active policies need timely retention review, and what evidence supports the choice?

What would the team need to know before contacting Ana or Ben?

Today's learning outcomes

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By the end of this meeting, you can:

1. Explain insurance roles, protection and the retention decision.
2. Ask stakeholder questions that clarify the outcome, action, constraints, and success criteria.
3. Write a project charter and connect requirements to analytical evidence.

The data science life cycle

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Life cycle focus

Business understanding and requirements

1. Understand the data and prepare usable histories.
2. Explore retention patterns and build a simple model.
3. Evaluate the evidence and communicate recommendations.
4. Deliver a reproducible portfolio and monitoring plan.

Requirements evolve: revisit them when the data reveals limits.

Protection and risk pooling

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- Insurance provides financial protection for covered events under a contract.
- Premiums from many policyholders help fund covered claims.
- Pooling spreads financial risk. It does not prevent the event.
- Life insurance supports beneficiaries after the insured person dies. Health and property insurance cover other specified risks.

T001: contract roles and protection

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- Fictional Harbor Life is the insurer. Alex owns T001 and is the insured person. Sam is the beneficiary.
- T001 has a 10-year term, a 100,000-unit death benefit and a 50-unit monthly premium.
- T001 has no cash or surrender value.
- Partner check: who owns the contract, whose life is covered, and who receives the covered death benefit?

Business requirements

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Concept

A business requirement describes what a user needs to decide or achieve, under stated constraints.

User	A retention supervisor reviews the monthly list.
Decision	Which eligible policies need review first?
Action	An agent checks a payment issue or explains approved options.
Constraint	Contact must occur early enough for useful support.

The requirements process

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1. **Ask:** clarify the decision and operational context.
2. **Record:** separate confirmed requirements from assumptions.
3. **Translate:** specify an analytical question and metric.
4. **Check:** confirm the interpretation with the stakeholder.

Interview follow-up

"We understand that you need a monthly ranked list. How many policies can your team review, and how early can you act?"

A requirement with a measurable limit

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Illustrative requirement, pending confirmation

The retention manager reviews a ranked list of active policies each month. Suppose the list has **100 policy slots**. Agree how multiple policies per customer affect contact capacity.

- Analytical question: which policies face lapse within the next three months?
- Evaluation: precision and recall among the top 100 eligible policies.
- Open question: which support actions can the team offer?

Quick quiz 1

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Question

Which statement specifies a business requirement?

1. Use a shallow decision tree.
2. Provide a monthly list of 100 active policies for retention review.
3. Create six colorful charts.

Answer and explanation

Quick quiz 1

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Question

Which statement specifies a business requirement?

1. Use a shallow decision tree.
2. Provide a monthly list of 100 active policies for retention review.
3. Create six colorful charts.

Answer and explanation

Statement 2 identifies an output, a decision context, and a capacity limit. Confirm the user, timing, and available support next.

Policy retention and customer retention

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Unit of analysis

A policy and a customer are different observation units. One customer can hold several policies.

Insurance example

Carlo holds a term life policy and a whole life policy. The term policy lapses. The whole life policy remains active.

- Policy retention: one policy remains active out of two.
- Customer retention: the customer still holds an active policy.

Confirm whether management cares about policies, customers, or premium income.

The lapse definition

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Operational definition

Specify the event rule, event date, and eligible population so another analyst can reproduce the outcome.

- Ask how unpaid premiums and grace periods affect policy status.
- Separate lapse from surrender, term expiry, maturity and death claims.
- Ask how reinstatement changes the recorded history.

Week 2 task: record the agreed definition and unresolved rules.

T001: coverage at the decision date

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- Simulation Year 2: premium due on 1 June. Assumed grace period: 2 June–1 July inclusive.
- On 30 June, T001 remains active during the grace period even though the premium is unpaid.
- Alternative A: full payment on 20 June keeps it active.
- Alternative B: still unpaid after 1 July, it lapses on 2 July. Later payment requires separate reinstatement approval.

Decision date and prediction horizon

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Illustrative scoring schedule

At the end of January, rank policies active on that date. Predict lapse during February, March, and April.

- A failed debit recorded on January 25 can inform the list.
- A cancellation request received on February 10 cannot.
- Ask whether agents have enough time to contact the selected policies.

Later evaluation: observe the complete outcome window before labeling a policy as a non-lapse.

An ambiguous target changes the answer

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“Predict every policy that closes”

This target combines exits with different causes and different possibilities for retention support.

Insurance example

A policy reaches its scheduled maturity. Counting it as lapse would change the question the model answers.

Better requirement: identify the agreed lapse event and document how other exits affect eligibility.

Quick quiz 2

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Question

One of a customer's two policies lapses. The other remains active. Which conclusion follows?

1. The insurer has lost the customer.
2. A policy has lapsed, while the customer still has an active policy.

Answer and explanation

Quick quiz 2

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Question

One of a customer's two policies lapses. The other remains active. Which conclusion follows?

1. The insurer has lost the customer.
2. A policy has lapsed, while the customer still has an active policy.

Answer and explanation

Statement 2. Policy lapse does not automatically mean customer loss. The requirement must specify which unit matters.

Success at the team's outreach capacity

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Suppose a test list contains 100 policies, of which 30 later lapse. Across all eligible policies, 60 later lapse.

$$\text{Precision at } 100 = 30/100 = 30\%$$
$$\text{Recall at } 100 = 30/60 = 50\%$$

- Precision: what proportion of selected policies later lapse?
- Recall: what proportion of all subsequent lapses appears in the list?

Illustrative numbers only. Agree on capacity and acceptance criteria with the manager.

Prediction and outreach benefit

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Concept

Elevated lapse risk tells us who may lapse. It does not establish who will benefit from contact.

- Ana may need help updating her payment details.
- Ben may need a review of options allowed by his policy.
- A pilot compares outreach with usual service to measure benefit.

Synthetic teaching data demonstrates analytical methods. Its patterns do not establish real customer behavior.

The 70-minute project mission

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Project task

In teams of 3–5, interview the retention manager and draft your project charter.

- **10 min** Plan questions and assign team roles.
- **20 min** Interview the manager and record answers.
- **30 min** Write the charter and requirements table.
- **10 min** Peer review and progress checkpoint.

Mark unknowns explicitly and name the person who can resolve them.

Week 2 deliverables and acceptance

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Submit: a charter with a labeled policy example, interview notes and requirements table.

- The charter names the decision, action and fictional policy roles.
- Outcome rules distinguish lapse from other policy exits.
- Timing and capacity have confirmed values or named open questions.
- Requirements connect to analytical questions and evidence.
- Assumptions and unresolved issues have owners and next steps.

Peer check: could another team explain your intended decision?

Three takeaways and next week

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1. A useful analysis begins with a stakeholder's decision.
2. Retention definitions determine the population and outcome.
3. Measurable requirements connect the question to evidence.

Week 3: data acquisition and understanding

Inspect the supplied synthetic tables, map fields to requirements, and assess whether the data can answer your questions.

Study example: a missed payment

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Fictional case

Ana's bank rejects her January premium debit. Her policy still appears active on the decision date.

Question	Is this a payment problem the team can help resolve?
Evidence	Debit result, earlier payments, status and contact history.
Action	An agent confirms the issue and explains the payment process.
Outcome	Track payment and policy status through the agreed horizon.

Rule to confirm: a failed debit alone does not define lapse.

Study example: a contact list agents can use

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Illustrative entry

Policy P104 ranks 12th among eligible policies. The latest debit failed and the policy remains active at the January 31 cutoff.

Agent needs Policy ID, rank, evidence date and assigned reviewer.

Eligibility Check current status, contact permission and recent outreach.

Next step Confirm the payment issue before offering approved support.

Record Contact attempt, customer response and follow-up date.

Open requirement: who refreshes eligibility between scoring and contact?

Study example: outreach benefit

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- Fictional randomized pilot: 200 policies from 200 different customers, 100 per group.
- Three-month lapse: outreach 18/100; usual service 25/100.
- Observed reduction: 7 percentage points. Assess uncertainty, costs and customer experience.
- For customers with several policies, keep all policies in the same assigned group.

Study reference: charter and scope

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- Problem: why does the retention team need the analysis?
- User and decision: who reviews the output and what choice follows?
- Scope: eligible policies, agreed lapse event and time window.
- Policy example: label fictional roles, terms and continuity rules.

Study reference: charter acceptance

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- Output: specify the report or review list.
- Success: link acceptance criteria to evidence.
- Constraints: capacity, support options, privacy and data limits.
- Open questions: record assumptions, owners and next review dates.

Study reference: requirements traceability

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Column	Illustrative entry
Requirement	Prioritize limited monthly outreach.
Analytical question	Which active policies rank highest for lapse within three months?
Metric or criterion	Precision and recall at top K, with K agreed by the stakeholder.
Supporting artifact	Baseline comparison and outreach prioritization table.
Status	Pending confirmation of capacity and acceptance criteria.

Update the same table as evidence emerges throughout the course.